Submission Declarations — Electrical Engineering (Archiv für Elektrotechnik, Springer Nature)

Manuscript Title: PSO-TCWavKAN: A Swarm-Optimized Temporal Convolutional Wavelet Kolmogorov-Arnold Network with Multi-Kernel Trend Decomposition and Asymptotic Linear Lower-Bound Guarantees for Resilient Power Transformer Thermal Dynamics Forecasting

Author: Nguyen Phu Toan

Affiliation: Department of Information Technology, Hanoi University of Business and Technology, 29A, Lane 124 Vinh Tuy Street, Vinh Tuy Ward, Hai Ba Trung District, Hanoi, Vietnam

Corresponding Author: Nguyen Phu Toan (Email: nphu211206@gmail.com | ORCID: 0009-0005-0529-2131)

Statements and Declarations

Availability of data and materials: The ETTh1 and ETTh2 benchmark datasets used in this study are publicly available components of the canonical Electricity Transformer Temperature benchmark established by Zhou et al. (Informer repository: https://github.com/zhouhaoyi/ETDataset). No private, proprietary, or confidential data were used. The public GitHub repository (https://github.com/nphu211206/KAN-PSO-Transformer-Forecasting) archives the complete v9.1 model code, aggregate benchmark and ablation CSV matrices, statistical reproduction scripts, parameter-count verification scripts, and publication figures under the MIT license. Per-seed raw training logs, model checkpoints, per-seed noise trajectories, and profiling logs are available from the author upon reasonable request.

Code availability: The complete software implementation of PSO-TCWavKAN v9.1, benchmark reproduction scripts, and figure plotting tools are publicly available at GitHub: https://github.com/nphu211206/KAN-PSO-Transformer-Forecasting under the open-source MIT License.

Competing interests: The author declares that there are no financial, professional, personal, or non-financial competing interests directly or indirectly related to the work submitted for publication.

Funding: The author received no specific grant or external financial support from any funding agency in the public, commercial, or not-for-profit sectors for this research.

Acknowledgements: Computational resources for benchmark evaluations and ablation training runs were provided by Kaggle Cloud GPU environments (NVIDIA Tesla T4). The author expresses gratitude to the open-source scientific Python community (PyTorch, NumPy, SciPy, pandas, Matplotlib) for developing foundational tools.

Ethics approval and consent to participate: Not applicable. This study utilizes open-source physical sensor telemetry and involves no human participants, tissue samples, or animal experiments.

Consent for publication: Not applicable. The manuscript contains no individual person's personal identifying data, images, or details in any form.

Author Contributions

Nguyen Phu Toan (NPT): conceptualization, mathematical formulation, software implementation, experiment execution, formal analysis, validation, data curation, visualization, writing – original draft preparation, writing – review and editing, and project administration. The author read and approved the final manuscript.